

JSS2 Mathematics Diagnostic Assessment (Part 1)

Curriculum: Cambridge Lower Secondary (Stages 7 & 8) & Nigerian JSS2
Cluster 1: Number Sense, Ratio, and Proportional Reasoning (Questions 1 - 20)

Mathematics - Cluster 1

No.	Question & Options	Ans	Diagnostic Focus / Concept Tag
1	Simplify the ratio $1.5\text{ kg} : 500\text{ g}$. A) $3:1$ B) $3:10$ C) $1:3$ D) $30:1$	A	Units conversion within ratios (Critical for Physics/ Chem).
2	Decrease 240 by 15%. A) 204 B) 225 C) 200 D) 210	A	Percentage decrease (Business Math foundations).
3	Express 0.00045 in Standard Form (Scientific Notation). A) 4.5×10^4 B) 4.5×10^{-4} C) 45×10^{-5} D) 0.45×10^{-3}	B	Indices and standard form (Required for JSS3 Science).
4	Divide 3500 items in the ratio $2:3:5$. What is the middle share? A) 700 B) 1050 C) 1750 D) 350	B	Proportional sharing (Multi-part ratios).
5	If 8 men can build a wall in 12 days, how many days will it take 6 men working at the same rate? A) 9 days B) 16 days C) 10 days D) 15 days	B	Inverse Proportion (High-level reasoning).
6	Solve: $(-12) \times (-3) - (+15) \div (-3)$. A) 21 B) 41 C) 31 D) 39	B	Directed numbers and BODMAS (Foundational accuracy).
7	Find the Highest Common Factor (HCF) of 48, 72, and 120. A) 12 B) 18 C) 24 D) 6	C	Prime factorization and HCF.
8	A map has a scale of $1:50,000$. If the distance between two towns on the map is 4cm, find the actual distance in kilometers. A) 2km B) 20km C) 200km D) 0.2km	A	Scales and Proportions (Real-world Geography/ Math link).
9	Find the value of $2^3 \times 3^2 \times 5^0$. A) 72 B) 36 C) 0 D) 120	A	Laws of Indices ($a^0=1$ test).
10	Convert $\frac{7}{8}$ to a percentage. A) 87.5% B) 70% C) 78% D) 80.5%	A	Fraction-to-Percentage conversion.
11	Round 45.678 to two significant figures. A) 45 B) 46 C) 45.68 D) 45.7	B	Significant figures vs Decimals (Precision check).
12	A car uses 5 liters of petrol for every 40km. How much petrol is needed for 200km? A) 20L B) 25L C) 15L D) 30L	B	Direct Proportion application.
13	Evaluate $\sqrt{1.44} + \sqrt[3]{0.008}$. A) 1.4 B) 1.22 C) 1.2 D) 1.02	A	Roots of decimals (Estimation and exactness).
14	The price of a phone rose from N50,000 to N62,500. Find the percentage increase. A) 20% B) 25% C) 15% D) 12.5%	B	Percentage change calculation.
15	Which is the correct arrangement in ascending order? A) $\frac{1}{2}$, 45%, 0.6 B) 45% , $\frac{1}{2}$, 0.6 C) 0.6, $\frac{1}{2}$, 45% D) 45% , 0.6, $\frac{1}{2}$	B	Comparing fractions, decimals, and percentages.
16	Find the missing term in the proportional set: $4:9 = 28:x$. A) 63 B) 56 C) 72 D) 45	A	Equivalent ratios.

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17	Estimate the value of $\frac{4.9 \times 20.1}{0.48}$ to the nearest whole number. A) 200 B) 20 C) 2000 D) 2	A	Approximation and magnitude checking.
18	Express 45 minutes as a fraction of 2 hours in simplest form. A) $\frac{3}{8}$ B) $\frac{2}{3}$ C) $\frac{3}{4}$ D) $\frac{1}{4}$	A	Time-based proportions.
19	Calculate 12.5% of \$800. A) 100 B) 80 C) 125 D) 90	A	Application of percentages.
20	In a school of 1200 students, the ratio of boys to girls is 5:7\$. How many more girls are there than boys? A) 200 B) 500 C) 700 D) 100	A	Comparative ratio analysis.